Pantothenate and Coenzyme A Biosynthesis: A Mechanistic and Evolutionary Review

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1. Executive summary

Coenzyme A (CoA) is a universal acyl-group carrier required for the tricarboxylic acid cycle, fatty-acid and polyketide metabolism, and protein acylation. Its biosynthesis is conventionally described as an eight-reaction linear pathway with two clearly separable sub-modules: (i) a **pantoate/pantothenate branch** (PanB → PanE-like reductase → PanC, fed by a β-alanine supply) that assembles the vitamin **pantothenate (vitamin B5)**, and (ii) a **five-step CoA-assembly branch** (PanK → PPCS/CoaB → PPCDC/CoaC → PPAT/CoaD → DPCK/CoaE) that converts pantothenate into CoA. The chemistry of the pathway is largely invariant across the tree of life, but the **protein architecture is strikingly plastic**: the first committed CoA step (pantothenate kinase) is catalysed by three unrelated enzyme families (types I/II/III), the ketopantoate-reductase step is served by at least three non-homologous enzymes (PanE, IlvC, PanG), consecutive activities are fused in some lineages (CoaBC; the bifunctional CoA synthase COASY), and the upstream β-alanine supply is delivered by several non-orthologous routes (PanD, pyrimidine degradation, promiscuous diamine metabolism, or uptake). The plasticity extends even to reaction order: **most archaea abolish free pantothenate as an intermediate**, phosphorylating pantoate first (pantoate kinase, PoK) and then condensing 4-phosphopantoate with β-alanine (phosphopantothenate synthetase, PPS) to reach 4′-phosphopantothenate by a route that inverts the canonical "condense-then-phosphorylate" logic
(P 24638914 ;

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CoA itself is an ancient cofactor traceable to the last universal common ancestor (LUCA), so the pathway represents a deeply conserved metabolic core onto which lineage-specific enzyme replacements, fusions, losses, and step-reorderings have been layered. The pathway is regulated principally at two post-translational nodes — feedback inhibition of PanK by CoA/acyl-CoA thioesters, and control of β-alanine output by the PanD–PanZ·acetyl-CoA complex — and is a validated antibacterial, antifungal, and antiparasitic drug target, most notably through pantothenamide antimetabolites. In humans, partial loss of PANK2 or COASY causes tissue-restricted neurodegeneration with brain iron accumulation (PKAN, CoPAN), a still-unexplained tissue selectivity that is among the field's key open questions.

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2. Definition and biological boundaries

What is inside the module. The system comprises the eight enzymatic conversions from 3-methyl-2-oxobutanoate (α-ketoisovalerate, a valine-pathway intermediate) and β-alanine to CoA:

1. 3-methyl-2-oxobutanoate → 2-dehydropantoate (ketopantoate) — **PanB**, ketopantoate hydroxymethyltransferase (KPHMT)
2. 2-dehydropantoate → (R)-pantoate — **PanE-like ketopantoate reductase (KPR)**
3. (R)-pantoate + β-alanine → pantothenate — **PanC**, pantothenate synthetase
4. pantothenate → 4′-phosphopantothenate — **PanK**, pantothenate kinase
5. 4′-phosphopantothenate + cysteine → 4′-phosphopantothenoylcysteine — **PPCS/CoaB**
6. 4′-phosphopantothenoylcysteine → 4′-phosphopantetheine — **PPCDC/CoaC**
7. 4′-phosphopantetheine → dephospho-CoA — **PPAT/CoaD**
8. dephospho-CoA → CoA — **DPCK/CoaE**

A useful conceptual division is between the **pantothenate-synthesis sub-module** (steps 1–3 plus β-alanine supply), which is restricted to bacteria, fungi, plants and some protists, and the **CoA-assembly sub-module** (steps 4–8), which is universal because animals and many parasites that cannot make pantothenate still must build CoA from salvaged vitamin B5 (P 34969059; P 38871981).

What is adjacent but should be treated separately. Several processes are routinely conflated with the pathway:

- **Phosphopantetheine handoff to carrier proteins.** The 4′-phosphopantetheine prosthetic arm of acyl carrier protein (ACP) and peptidyl/polyketide carrier proteins is installed not by the CoA pathway but by **Sfp-/AcpS-type phosphopantetheinyl transferases (PPTases)**, which transfer the phosphopantetheine moiety from CoA onto a conserved serine of apo-carrier proteins (P 36618868). This is a downstream, CoA-consuming reaction, not a biosynthetic step.
- **CoA salvage.** Extracellular CoA/dephospho-CoA and pantetheine can be degraded and re-imported; salvage feeds the same pool but uses transporters and ectoenzymes outside the de novo module (P 34969059).
- **Downstream CoA-dependent metabolism** (acyl-CoA thioester formation, acetylation, etc.) and **β-alanine's non-CoA fates** are out of scope.
- **The upstream branch-point.** Step 1 draws its carbon skeleton from α-ketoisovalerate, shared with branched-chain (valine) amino-acid biosynthesis, and its hydroxymethyl group from **5,10-methylenetetrahydrofolate**, tying the pathway to one-carbon/folate metabolism (PMID 6463). These are supplying pathways, not part of the module.

Competing definitions. Textbooks often present a fixed "universal five-step" CoA route with named *E. coli* genes. This is an oversimplification: the number of polypeptides (fusions), the identity of the enzymes (non-orthologous displacement), the nucleotide cofactor (CTP vs ATP at step 5), the ordering of phosphorylation vs β-alanine condensation (canonical vs archaeal), and the very presence of the pantothenate branch all vary. Even the intermediate set differs — in most archaea free pantothenate is never formed (Section 5). The most defensible definition is therefore **reaction-outcome-based** (conversion of pantoate + β-alanine + cysteine, via 4′-phosphopantothenate, to CoA) rather than **gene-based** or tied to a fixed intermediate list.

3. Mechanistic overview

Obligatory linear order. The pathway is a strict metabolic assembly line: each product is the sole substrate of the next enzyme, so the sequence 1→8 is obligatory for de novo synthesis. The only branch is the **entry of β-alanine at step 3**, which is supplied in parallel and can originate from several routes (Section 5). Steps 1–3 are conditional/accessory in organisms that salvage pantothenate; steps 4–8 are obligatory wherever CoA is made from pantothenate.

Step 1 — PanB (KPHMT). A folate-dependent aldol-type transfer: the hydroxymethyl group of 5,10-methylenetetrahydrofolate is added to C3 of α-ketoisovalerate to give ketopantoate, with Mg^{2+} orienting the substrate for deprotonation (P 12906829; PMID 6463). PanB is a **$(\beta\alpha)_8$ TIM-barrel of the phosphoenolpyruvate/pyruvate superfamily** that assembles into a decamer via C-terminal helix swapping (P 12842039; P 12837791).

Step 2 — ketopantoate reductase. NAD(P)H-dependent stereospecific reduction of the C2 keto group to give **(R)-pantoate**. Canonical PanE is a monomeric, NADPH-preferring enzyme of the 6-phosphogluconate-dehydrogenase C-terminal-domain superfamily with a Rossmann N-terminal domain; catalysis requires an induced-fit domain closure over the nicotinamide cofactor (P 11724562; P 31136784).

Step 3 — PanC (pantothenate synthetase). An ATP-dependent amide ligation of (R)-pantoate and β-alanine. The mechanism is a well-characterised **Bi-Uni-Uni-Bi ping-pong**: ATP binds first, then pantoate; PPi is released as a **pantoyl-adenylate intermediate** forms; β-alanine then binds and attacks, releasing pantothenate and AMP (P 11669627). The pantoyl-adenylate was confirmed by ^{31}P NMR and shown to be kinetically competent.

Step 4 — PanK (pantothenate kinase). ATP-dependent phosphorylation of the terminal hydroxyl of pantothenate to 4′-phosphopantothenate. This is the **first committed and generally rate-limiting step**, and the major regulatory node (P 40754168; P 38871981).

Archaeal alternative to steps 3–4. Most archaea lack both PanC and PanK and instead reach the shared intermediate 4′-phosphopantothenate by a reordered two-enzyme route: **pantoate kinase (PoK, COG1829)** phosphorylates (R)-pantoate to 4-phosphopantoate, and **phosphopantothenate synthetase (PPS, COG1701)** then performs ATP-dependent condensation of 4-phosphopantoate with β-alanine (via a phosphopantoyl-adenylate intermediate) to give 4′-phosphopantothenate directly (P 24638914; P 22940806; P 23200110). Free pantothenate is therefore never formed in these organisms; phosphorylation precedes amide-bond formation, the reverse of the canonical order.

Step 5 — PPCS/CoaB. Nucleotide-dependent ligation of cysteine to 4′-phosphopantothenate via an **acyl-nucleotidyl intermediate**. Bacterial enzymes are **CTP-dependent** and proceed through a 4′-phosphopantothenoyl-CMP intermediate; eukaryotic PPCS is **ATP-dependent** (P 15530362; P 12140293).

Step 6 — PPCDC/CoaC. FMN-dependent oxidative decarboxylation of the cysteinyl carboxylate to give the thioethanolamine (phosphopantetheine). It proceeds through an oxidatively decarboxylated **aminoenethiol intermediate** that is then re-reduced by a redox-active active-site cysteine — mechanistically distinct from PLP-dependent decarboxylation (P 11358972; P 11923307).

Step 7 — PPAT/CoaD. Reversible adenylyl transfer from ATP to the phosphate of 4′-phosphopantetheine, forming dephospho-CoA and PPi. Bacterial CoAD is a hexameric member of the nucleotidyltransferase (α/β phosphodiesterase) superfamily and is a feedback-controlled regulatory node in bacteria (P 38456905; P 29551072).

Step 8 — DPCK/CoaE. ATP-dependent phosphorylation of the 3′-OH of the dephospho-CoA ribose to yield CoA. CoaE is a **P-loop NTPase** with CoA-domain, nucleotide-binding, and lid subdomains (P 21264299; P 21731728).

4. Major molecular players and active assemblies

Step	Activity	Representative enzyme(s)	Fold / assembly	Cofactor / co-substrate	Key refs (PMID)
1	KPHMT	PanB	$(\beta\alpha)_8$ barrel, decamer; PEP/pyruvate superfamily	5,10-CH_2-THF, Mg^{2+}	12906829, 12842039, 6463
2	KPR	PanE; IlvC (moonlight); PanG	6-PGDH-like (PanE); KARI (IlvC); distinct (PanG)	NADPH/ NADH	11724562, 29791499, 23243306
3	Pantothenate synthetase	PanC (canonical); PoK+PPS (archaea, steps 3–4 combined)	Rossmann-like nucleotide-binding N-domain; homodimer	ATP → pantoyl-adenylate	11669627, 17932772, 24638914
4	Pantothenate kinase	Type I (CoaA), Type II (PanK1–4), Type III (CoaX)	actin/ASKHA (I, II); ASKHA (III)	ATP	18186650, 17631502, 24784177
5	PPCS (CoaB)	Standalone PPCS; CoaB domain of CoaBC	dinucleotide-binding fold; dimer	CTP (bact.) / ATP (euk.)	15530362, 12140293
6	PPCDC (CoaC)	Standalone PPCDC; CoaC domain of CoaBC; AtHAL3a	HFCD flavoprotein, trimer	FMN	11358972, 11923307
7	PPAT (CoaD)	Bacterial CoaD; PPAT domain of COASY	nucleotidyltransferase, hexamer (bact.)	ATP	38456905, 11980892
8	DPCK (CoaE)	Bacterial CoaE; DPCK domain of COASY	P-loop NTPase	ATP	21264299, 11923312

Architectural packaging varies by lineage. In most bacteria steps 5 and 6 are fused into a single bifunctional **CoaBC (Dfp)** flavoprotein (N-terminal decarboxylase CoaC + C-terminal ligase CoaB) (P 12140293), and steps 7 and 8 are catalysed by separate CoaD and CoaE. In higher eukaryotes the reverse fusion is seen: steps 5 and 6 are separate (PPCS, PPCDC), while steps 7 and 8 are fused into the bifunctional **CoA synthase (COASY, ~60 kDa)** carrying both PPAT and DPCK domains (P 11980892; P 11923312). The complete human enzyme set was defined by comparative genomics and reconstituted in vitro (P 11923312).

Table note: in most archaea, steps 3 and 4 are replaced by **pantoate kinase (PoK, COG1829)** + **phosphopantothenate synthetase (PPS, COG1701)**, which phosphorylate pantoate before condensing it with β-alanine and thus never form free pantothenate (structures: P 31697438 P 24638914 P 24021277; see Section 5).

β-Alanine supply enzyme. The canonical supplier, **PanD (L-aspartate-α-decarboxylase)**, is unusual: a tetrameric β-barrel translated as an inactive proenzyme that matures by **autocatalytic self-cleavage through an ester intermediate**, generating a catalytic **pyruvoyl** group (not PLP) (P 9546220; P 30073608).

5. Evolutionary and cell-biological variation

Three unrelated pantothenate-kinase families (step 4). Type I (e.g., *E. coli* CoaA), Type II (predominantly eukaryotic PanK, four human isoforms PANK1–4), and Type III (CoaX) are **distinct in sequence, fold and kinetics**; Type III belongs to the ASKHA/acetate-and-sugar-kinase superfamily, has an unusually high K_m for ATP, is resistant to CoA feedback inhibition, and cannot activate pantothenamide analogues (P 18186650).

This is a textbook case of **non-orthologous gene displacement**, and some bacteria (*B. subtilis*, *M. tuberculosis*) encode both type I and III enzymes with type I contributing the larger share of the CoA pool (P 24784177).

Which PanK is essential is organism-dependent: type III CoaX is essential in *Bacillus anthracis* (P 18641144) but dispensable in *M. tuberculosis*, where type I CoaA is essential (P 20576686) — a well-documented cross-organism discrepancy.

Multiple reductases (step 2). Canonical **PanE**, the moonlighting ketol-acid reductoisomerase **IlvC** (which rescues *panE* loss in *Zymomonas* and *Salmonella*; P 29791499), and the non-homologous **PanG** (sole KPR in *Francisella*, with homologs in *Enterococcus*, *Coxiella*, *Clostridioides*; P 23243306) independently perform the same reduction. Genome surveys show pantothenate-pathway genes are frequently duplicated and horizontally transferred, and *panE*/*panD* are frequently lost or replaced (P 31136784).

Archaeal reordering of steps 3–4 (a distinct route to 4′-phosphopantothenate). Whereas bacteria and eukaryotes make free pantothenate (PanC) and then phosphorylate it (PanK), most archaea encode neither enzyme and instead use **pantoate kinase (PoK)** followed by

phosphopantothenate synthetase (PPS), phosphorylating pantoate *before* condensing it with β-alanine (24638914; 22940806). Both enzymes are structurally characterized (PoK and PPS from *Thermococcus kodakarensis* / *T. onnurineus*; 31697438; 24021277) and their homologs are widely conserved across archaeal genomes (23200110). Regulation also differs: archaeal PPS is not feedback-inhibited by CoA/acetyl-CoA, while PoK shows product inhibition by 4-phosphopantoate (22940806) — so the CoA-feedback node has shifted relative to the PanK-centred control of bacteria/eukaryotes. This is the deepest example in the pathway of achieving the same outcome (4′-phosphopantothenate) by different molecular means *and* in a different order.

Variable β-alanine routes. PanD-dependent aspartate decarboxylation is canonical, but β-alanine is also produced by **β-alanine synthase of the reductive uracil-degradation (pyrimidine catabolism) pathway** (*Rhizobium etli*; 32112625) and by a **promiscuous route via 2,4-diaminobutyrate and 1,3-diaminopropane** (*Acinetobacter baylyi*; 35623386). Loss of *panD* alone renders *Zymomonas mobilis* pantothenate-auxotrophic (28655181).

Compartmentalization. In plants, KPHMT is mitochondrial while pantothenate synthetase is cytosolic, requiring intermediate transporters; plants also lack a recognizable PanD, implying an alternative β-alanine source

(P 14675432).

In humans, PANK2 acquired a **primate-specific mitochondrial targeting signal** (mouse PanK2 is cytosolic), an important caveat when modelling human PANK2 disease in mice (P 17825826).

The four human PanK isoforms differ in tissue expression, subcellular localization, and feedback sensitivity (PanK3 is the most acetyl-CoA-sensitive, $IC_{50} \approx 1$ μM; P 16040613 ; P 12095677).

Deep origin. CoA is among the cofactors inferred to have been present in **LUCA** (P 27562259), and CoA metabolism is under strong purifying selection in deep-branching methanogens (Tajima's $D \approx -2.86$; P 40051064). The pathway is therefore ancient, with the PanK family expansions, enzyme fusions, and non-orthologous replacements representing **later, lineage-specific elaborations** on a conserved chemical core. For the universal CoA-assembly branch, the standalone bacterial CoaD/CoaE most closely mirror the minimal ancestral gene set and are the most informative single-domain representatives, whereas the fused eukaryotic COASY is derived. For the pantothenate-to-4′-phosphopantothenate conversion, however, the ancestral state is genuinely **unresolved**: the bacterial/eukaryotic PanC+PanK route and the archaeal PoK+PPS route are non-homologous, so it is not clear which (if either) was present in LUCA, and the deep split between them cautions against treating any one of Type I PanK, Type III CoaX, or archaeal PoK as "the" ancestral kinase.

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6. Constraints, dependencies, and failure modes

Order constraints — and where they break down. Within the CoA-assembly branch the order is genuinely fixed: cysteine must be attached (step 5) **before** decarboxylation (step 6), and adenylylation (step 7) precedes 3′-phosphorylation (step 8), because each intermediate is the unique substrate of the next enzyme. By contrast, the ordering of *pantothenate* *condensation vs its phosphorylation* is **not** fixed across life: bacteria/eukaryotes condense pantoate + β-alanine first (PanC) and phosphorylate the product (PanK), whereas most archaea phosphorylate pantoate first (PoK) and then condense (PPS), never forming free pantothenate (P 24638914; P 22940806). This archaeal route directly rules out the intuitive assumption that free pantothenate is an obligatory universal intermediate. β-Alanine must be available before the condensation step (step 3 / PPS) can complete, regardless of route.

Substrate/cofactor constraints. Step 1 cannot proceed without folate-derived one-carbon units; step 3 requires ATP and forms an obligatory pantoyl-adenylate; bacterial step 5 specifically requires CTP whereas eukaryotic step 5 requires ATP — a nucleotide specificity that rules out simple interchange of bacterial and eukaryotic PPCS in heterologous systems (P 15530362). Step 6 requires FMN and a redox-active cysteine; loss of the oxidative machinery blocks decarboxylation.

Regulatory nodes (flux control is post-translational, not transcriptional). (i) **PanK** is feedback-inhibited by CoA and acyl-CoA thioesters, coupling CoA synthesis to demand (P 16040613; P 17631502). (ii) **β-Alanine supply** is throttled by the **PanD–PanZ·acetyl-CoA complex**, in which PanZ binding sequesters PanD's pyruvoyl cofactor as a ketone hydrate when CoA/acetyl-CoA is abundant (P 28832133). (iii) In bacteria, **CoaD/PPAT** is additionally feedback-regulated, and mycobacterial **CoaE** is inhibited by CTP and modulated by monomer–trimer equilibrium (P 21731728).

Failure modes. All eight activities are individually essential where de novo synthesis operates: *panC* and *panK* are essential in *M. tuberculosis* (P 22840772;

(P 20576686), and *coaD*/*coaE* are essential despite earlier annotation ambiguity (P 22954585). In humans, **partial** loss of PANK2 causes PKAN and partial loss of COASY causes CoPAN — both **neurodegeneration with brain iron accumulation (NBIA)** with globus-pallidus iron deposits (P 30740736) — while **near-complete** COASY loss is prenatally lethal (pontocerebellar hypoplasia, microcephaly, arthrogryposis; P 30089828). Drosophila models link CoASY decline to impaired mitochondrial integrity and reduced complex I/III activity (P 39985665). Conversely, PanK/CoA **excess** is also deleterious: transgenic PANK2 overexpression elevates muscle CoA and impairs performance (P 28189602).

Therapeutic exploitation. The pathway is a validated drug target in bacteria, fungi, and apicomplexan parasites (P 41914741; P 34969059; P 37762222). **Pantothenamides** are pantothenate analogues that are phosphorylated by PanK and elaborated into inactive CoA/ACP antimetabolites; their clinical development has been limited by serum **pantetheinase** degradation, motivating pantetheinase-resistant "inverted-amide" and bioisostere analogues with Gram-positive and antiplasmodial activity (P 31171848; P 29332383; P 40801835).

Because *Plasmodium* salvages pantothenate and depends on CoA assembly, steps 4–8 (but not 1–3) are the exploitable targets there (34969059).

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7. Controversies and open questions

1. **Tissue selectivity of human disease.** Why do PANK2 (PKAN) and COASY (CoPAN) defects, in a universally essential pathway, selectively injure the CNS and cause iron accumulation? Whether iron accumulation is primary or secondary remains debated, and the mechanistic link to CoA is unresolved (30740736).
2. **Which PanK is "the" essential kinase.** Essentiality of type I vs type III CoaX is genuinely organism-specific (essential type III in *B. anthracis*; essential type I in *M. tuberculosis*), and generalizing from one species is a recurring error (18641144; 20576686).
3. **β-Alanine supply is under-mapped.** The frequency and identity of non-PanD routes (pyrimidine degradation, promiscuous diamine metabolism, uptake) across taxa are incompletely catalogued; plant β-alanine sourcing in the absence of PanD is still not fully defined (14675432; 35623386).
4. **Regulation in eukaryotes.** Feedback control is best characterised in bacterial and mammalian PanK; whether CoaD/CoaE-equivalent nodes regulate flux through the fused COASY, and the physiological role of COASY's interaction with the P-body scaffold EDC4, remain open (22982864).
5. **Cross-organism data mixing.** Much mechanistic detail is drawn from *E. coli*, *M. tuberculosis*, and human isoforms; kinetic and structural parameters (e.g., NAD(P)H

preference of KPR, CTP vs ATP at step 5) are not safely transferable between lineages (P 31136784 ; P 15530362). The archaeal PoK/PPS route is a stark reminder that even the intermediate set and step order can differ, so "the CoA pathway" cannot be assumed identical across domains (P 24638914).

6. **Annotation reliability.** Non-orthologous displacement (PanE/IlvC/PanG; three PanK families) means genome annotations based on homology systematically miss or mis-assign pathway genes, complicating comparative and drug-target genomics (P 23243306).
7. **Ancestral route to 4′-phosphopantothenate.** Because the bacterial/eukaryotic (PanC+PanK) and archaeal (PoK+PPS) solutions are non-homologous, whether LUCA made free pantothenate at all — and which route is derived — remains an open evolutionary question (P 22940806).

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Prepared as an autonomous discovery-agent synthesis. Uncertainty is flagged in Sections 6–7; broad claims are anchored to the primary literature cited above, and organism-specific results are not generalized beyond their evidentiary basis.
